

Analyzing and Visualizing Query Results

Lesson objectives

Analyzing and Visualizing Query Results

1. List some of the client tool connections offered by Starburst
2. Compare the benefits of direct queries vs. importing data into client tools
3. Explain the preference for using Starburst-supported BI tools

Starburst Client Tools

Starburst connects with a wide range of client tools, enabling you to do far more than run queries. These tools let you build complex reports, create interactive charts, design dashboards, and generate many other valuable insights.



Working With BI Tools Connected to Starburst

Modes of Working

Importing Data Sets (In-Memory Reporting)



- Loads data into the BI tool for analysis
- **! Disadvantages:**
 - Data can become stale
 - High memory consumption

Direct Query (Recommended)



- Fetches fresh data directly from the source
- Supports real-time reporting
- Better performance & scalability

Starburst Advantage

- Eliminates need to import data sets
- Connects directly to multiple data sources
- Joins data efficiently across sources
- Provides a single point of access for live analytics

Key Takeaway: Use direct queries through Starburst for real-time insights and scalable analytics

Starburst Clients & BI Tool Support

Starburst-developed client connectors ensure control, faster fixes, and seamless user experience.

Starburst-supported Clients 🔌

✅ Maintained directly by Starburst

✅ Quick fixes and improvements

✅ Clear ownership for troubleshooting

Simplified Installation ⚙️

💻 Pre-installed in Tableau & Power BI

⚡ No manual setup required

Comprehensive Support 🛠️

🌐 Supports Desktop, Server, and SaaS (depending on tool)

NEW Clear guidance for each version

🐛 Starburst-supported bug fixes

ODBC Driver and Data Optimization

Best Practices for Performance

- ✅ Use ODBC Driver v3 to enable the [spooling protocol](#).
- ⚡ Aggregate in queries and pull only the data you need.
- ❌ Avoid extracting huge raw datasets for visualization.
- 🔄 Spooling enables parallel downloads, reducing transfer time from minutes to seconds for millions of rows.

Lesson summary

Analyzing & Visualizing Query Results

1. Starburst connects with a wide range of BI client tools (e.g., Tableau, Power BI, Jupyter, Alteryx, Qlik, DBeaver) for complex reporting, interactive charts, dashboards, and valuable insights.
2. The recommended method is direct query through Starburst, which provides real-time insights, fresh data, and enhanced scalability.
3. Starburst's direct query advantages include eliminating data imports, direct connection to multiple data sources, efficient data joining, and a single point of access for live analytics.
4. Starburst offers comprehensive support for its client connectors, ensuring quick fixes, simplified installation (pre-installed for some tools like Tableau & Power BI), and clear ownership for troubleshooting.
5. For optimal performance, use ODBC Driver v3 to enable spooling (for parallel, faster data downloads) and aggregate queries to pull only the necessary data, avoiding huge raw datasets.